

Quick Python guide and resources

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What is Python

Python is high-level programming language with numerous advantages:

- ▶ Simple and intuitive syntax;
- ▶ High-level (accomplish complex tasks with less coding);
- ▶ Many contributors, powerful packages online; Widely used in scientific computing and many computer science disciplines (especially machine learning or artificial intelligence);
- ▶ Free to use; Works on Linux/Mac/Windows.

How to start

Best way is to do the followings in order:

1. Read online tutorial (often faster) or videos – get a basic understanding and knowledge of syntax; (See the last slide for some good tutorials)
2. Read some code packages (that you know exactly how the algorithm works) written by more experienced programmers.
3. Try to write a program that implements an algorithm yourself;
4. When you have a question, google it – most of time you'll find an answer in seconds.

Setup

- ▶ If you use Linux/Mac, the OS should have Python (likely version 2.7) installed already. Type "python" in terminal to launch.
- ▶ If you use Windows and need to install Python, or you want other version, just go to Python's official website to download: <https://www.python.org>.
- ▶ It is recommended to use an IDE (integrated development environment) than coding in shell/terminal. Choice of IDE is more of personal taste. If you like a recommendation, you can start with spyder <https://pythonhosted.org/spyder/> which looks like MATLAB.
- ▶ For easiest start, you may just directly install anaconda which will manage IDE and many other packages for you: <https://www.anaconda.com/download>

Some basics

- ▶ Python functions and scripts are saved as “.py” files (like “.m” files for MATLAB); Sometimes Python will generate auxiliary “.pyc” files when you compile, you can ignore them.
- ▶ One .py file can contain multiple functions (in contrast one .m file in MATLAB is often implemented as one function). This file can be imported when you write a script, so that all functions included can be called by the script.
- ▶ To execute a script (say “my_script.py”), you can type “python my_script.py” in shell/terminal. If you use IDE, there will be a “run” button on top of your editor that directly execute your script (just like MATLAB).
- ▶ There are many powerful packages available. We will mostly use NumPy, SciPy, and Matplotlib.

Online tutorials

- ▶ Basic syntax
https://www.tutorialspoint.com/python/python_basic_syntax.htm
- ▶ Python NumPy Tutorial
<http://cs231n.github.io/python-numpy-tutorial/>
- ▶ NumPy for MATLAB users
<https://docs.scipy.org/doc/numpy-dev/user/numpy-for-matlab-users.html>
- ▶ More advanced tools in scientific computing–SciPy
<https://docs.scipy.org/doc/scipy/reference/index.html>
- ▶ Python can draw beautiful figures–“matplotlib” package
<https://matplotlib.org/tutorials/index.html>